

# Neuromorphic Human-Computer Interaction: A Wikipedia Study

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## ABSTRACT

I expect we write this section last. Lorem ipsum bla bla bla

## CCS CONCEPTS

• **Computer systems organization;**

## KEYWORDS

Neuromorphic Computing, Some more keywords

## 1 INTRODUCTION

This is the introduction section

## 2 BACKGROUND

In this section we provide an overview of background literature in current academic field.. Through this, we aim to embed our project in this field, and present the foundations of our research.

The idea that Wikipedia contains information on semantic relationships is not a new one. Several works investigate the possibility of extracting additional information on the semantic distance between concepts from Wikipedia, all of these show promising results[2–4]. These findings suggest a relationship between the semantic similarity of pages, and their distance in the Wikipedia graph-structure. [extend?]

The papers mentioned above mostly focus on information that can directly be extracted from the structure of data within Wikipedia. However, there are other aspects to consider as well, especially the way this Wikipedia data-structure is actually utilized by its users. West et al. [5] investigate the possibility of extracting semantic distances from a dataset of Wikispeedia games. This dataset consists of network traversal paths, where each path represents a person playing the Wikipedia game. In this game, a user is tasked to navigate from source-page A to goal-page B in the least amount of steps, thus these traces show the behaviour of targeted navigation on the Wikipedia network. This paper presents several interesting findings and insights, first and foremost that these traces produce semantic distances between concepts that are sensible when presented to human evaluators, improving over the LSA baseline. A second interesting observation they make is the fact that wayfinding for this task consists of two phases, a generalization phase and a specification phase. Since source-page A and goal-page B don't necessarily

have a clear human-sensible path between the two directly, players of the game often first navigate to a "hub"-page.

<https://link.springer.com/article/10.1007/s13218-015-0417-5>

LLMs can do wikipedia traversal. Why? Semantic distance indicative of distance? [5] Paper Ana -> idea that there is a relationship between time/clicks and semantic distance [10][9]

Wikipedia traversal overview [6, 7] + "However, individuals' performance when navigating a knowledge space is influenced by personal characteristics, with younger and multilingual individuals demonstrating higher navigation abilities (Zhu, Yasseri, & KertÄlsz, 2024)." + hypothesis to state that this suggests that internal bayesian might be closer to optimal path rather than semantic distance only -> younger people have more knowledge of the possible network (or think about it like this more)

Fits law -> expect somewhat of a similar relationship + log + the semantic/optimal distance as distance, width analogous to ease of access + Intuition because of the hub behaviour

LLM to find semantic distances: - why? - what llm?

Network traversal to determine semantic distance: <https://sci-hub.se/10.1037/xlm0000391>

Semantic distance as a predictor of click behaviour: <https://www.tandfonline.com> + tf-idf rather than title similarity - but not insane to use titles [8] + interesting note on structure

Might be interesting: <https://dl.acm.org/doi/pdf/10.1145/3488560.3498496> <https://arxiv.org/pdf/1203.1858>

Specifically, good to see people combine information from wikipedia into semantic distance already. New insights might be learned from user behaviour, and inversely we might learn more about human behaviour:

Some examples. A paginated journal article [1], an

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## A ADDITIONAL INFORMATION

This is the appendix